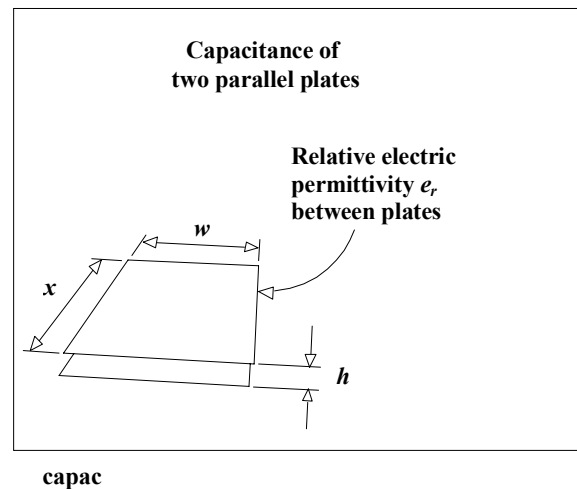


Formulas included in this spreadsheet:

Capacitance of two plates	CPLATE()
Impedance magnitude of capacitor at one frequency	XCF()
Impedance magnitude of capacitor as seen by rising edge	XCR()

Variables used:

w	Width of plate overlap (in.)
x	Length of plate overlap (in.)
h	Height of one plate above the other (in.)
er	Relative dielectric constant of material between plates



Capacitance of two plates (F):

$$\text{CPLATE}(w, x, h, \epsilon_r) := 2.249 \cdot 10^{-13} \cdot \frac{\epsilon_r \cdot x \cdot w}{h}$$

A power and ground plane separated by 0.010 in. of FR-4 dielectric ($\epsilon_r = 4.5$) share a capacitance of 100 pF/in.²

Halving the separation doubles the capacitance.

Impedance magnitude of capacitor at frequency f (Ω):

c	Capacitance (F)
f	Frequency (Hz)

$$\text{XCF}(c, f) := \frac{1}{2 \cdot \pi \cdot f \cdot c}$$

The impedance, at 100 MHz, of a
100-pF capacitor is 16 Ω .

$$XCF(100 \cdot 10^{-12}, 10^8) = 15.915$$

Impedance magnitude of capacitor as seen by rising edge (Ω):

c Capacitance (F)

tr 10-90% rise time (s)

$$XCR(c, tr) := \frac{tr}{\pi \cdot c}$$

The impedance, as seen by a 5-ns rising edge
of a 100-pF capacitor is 16 Ω .

$$XCR(100 \cdot 10^{-12}, 5 \cdot 10^{-9}) = 15.915$$